```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.widgets import Slider, Button
# x values
x = np.linspace(-5, 5, 200)
# Initial value of a
a = 1
# Create Figure and Axes
fig, ax = plt.subplots()
plt.subplots_adjust(bottom=0.3)
# Initial curve
line, = ax.plot(x, a*x**2)
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_title("Interactive y = a*x²")
# ----------------------
# Slider
# ----------------------
slider_ax = fig.add_axes([0.2, 0.12, 0.6, 0.04])
slider = Slider( slider_ax,  "a",  0.1,  5.0,  valinit=1)
# ----------------------
# Reset Button
# ----------------------
button_ax = fig.add_axes([0.4, 0.02, 0.2, 0.05])
reset_button = Button(button_ax,  "Reset")
# ----------------------
# Update function
# ----------------------
def update(val):
    a = slider.val
    line.set_ydata(a * x**2)
    fig.canvas.draw_idle()
slider.on_changed(update)
```

```
# ------------------------
# Reset function
# ------------------------
def reset(event):
    slider.reset()
reset_button.on_clicked(reset)
plt.show()
```

